Fiscal Year:	FY 2015	**Task Last Updated:**	FY 05/18/2017
PI Name:	Napolitano, Ralph E Ph.D.		
Project Title:	Three-Dimensional Structures and Interface Dynamics in Univariant and Invariant Eutectic Solidification in a Ternary Alloy		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
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Space Biology Element:	None		
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Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009-10 Materials Science Rapid Turn Around Flight Opportunity
Start Date:	10/01/2010	**End Date:**	09/30/2015
No. of Post Docs:		**No. of PhD Degrees:**	1
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:	Effective May 15, 2014, Project Leadership will transition to Professor Amber Genau (University of Alabama, Birmingham). Professor Napolitano will remain as a co-investigator and advisor to Professor Genau.		
COI Name (Institution):	Genau, Amber (University of Alabama, Birmingham)		
Grant/Contract No.:	NNX10AT61G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The diverse and expansive spectrum of growth morphologies that are attainable through solidification processing of ternary and higher order systems provides a promising avenue for accessing new structures, materials, and devices with enhanced functionality. However, the challenge of controlling the complex interplay between coupled-phase topology, local defect structures, and non-planar growth modes requires a more complete understanding of the mechanisms of pattern formation and the influence of applied external forces. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration, faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and applied thermal field.

SIGNIFICANCE

Multiphase multi-scale material structures offer great promise for discoveries of as yet unseen fundamental material responses, new types of device functionality, and improved levels of materials performance. Indeed, innovations in functional heterostructures have led to great developments in photovoltaics, photonics, and even metamaterials. Moreover, limitations to future applications of these and other multiphase materials arise from difficulties in exerting a high level of control over various structural features. As in situ or natural processing avenues to such materials continue to be explored, it is clear that multiphase solidification is one very important pathway toward these and other remarkable materials. The objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled multiphase structures through advanced solidification processing. Moreover, flight-based experiments are critical to both of these areas, serving (i) to provide an environment for fundamental measurement of dynamics that cannot be effectively observed in ground-based studies and (ii) as a potential environment for the fabrication of novel materials.

CRITICAL QUESTIONS:

• What are the geometric and topological principles governing the distribution of phases, fault domains, lamellar fault structure, grain structure, and interface morphology? Further, how do these depend on specific crystal orientations with respect to (i) neighboring lamellae, (ii) overall periodicity, (iii) the directional growth axis, and (iv) the thermal gradient and specimen geometry?

• How are the geometric relationships correlated with pattern formation dynamics and the selection of specific growth morphologies? Specifically, how do the basic structural and crystallographic features give rise to (i) lamellar fault formation and ordering, (ii) the maze-lamellae transition, (ii) eutectic cells and dendrites, and (iv) selection of 3-phase structures?

• How do the dynamics respond to thermal and geometric constraints that could be applied for the purpose of controlling the structure?

OVERARCHING PRINCIPAL OBJECTIVES:

• Establish an analytical framework for 3D quantitative description of the multiphase microstructures, combining geometric and crystallographic parameters for phase boundaries, domain boundaries (faults), grain boundaries, and the solid-liquid interface itself.

• Understand the local structural interactions and fundamental mechanisms that govern morphological dynamics and the evolution of complex 3D multiphase microstructures.

• Employ the newly developed quantitative framework and mechanistic understanding to investigate the collective dynamics of pattern formation, specifically examining steady-state pattern dynamics, the natural transient under directional growth conditions, the development of non-planar growth modes along the univariant path, and the response to thermal and geometric constraints.

• Utilize new theoretical understanding and predictive capability to (i) obtain a higher level of microstructural control, (ii) access new or more uniform growth structures, (iii) identify strategies to achieve new or better multiphase structures with novel or enhanced functionality. In addition to the control of alloy composition, growth velocity, and thermal gradient, avenues to ultimately explore will include novel thermal fields, geometric constraints, other applied fields, seeded growth, and techniques that require microgravity conditions.

SCOPE OF RESEARCH ACTIVITIES (includes both ground-based and microgravity experiments)

• Characterization and parameterization of the multi-scale 3D structure (phase distribution, interface structure, distribution of phase boundaries and orientation, eutectic fault distribution, domain structure, and larger scale grain structure);

• Quantification of natural directional growth transient (maze-lamellar transition, grain selection, fault formation and configuration, and fault domain geometry);

• Investigation of the response to forced thermal constraints (including off-axis thermal gradients that may be switched abruptly or rotated continuously, examining “locked” grain orientations and lock/unlock thresholds in 3D); and

• Investigation of response to forced geometric constraints (including off-axis growth from random and/or seed crystals, tilted and helical geometries, and geometric cross-sectional expansion or contraction).

COLLABORATION

While the proposed project is justifiable on its own merits, the work described here is also intended to provide collaborative support to the existing European Space Agency (ESA) project titled “Solidification along a Eutectic Path in Ternary Alloys” (SETA), which has made remarkable scientific progress in recent years. Facilitating open scientific interaction between the PI and the SETA team members (and between NASA and ESA), the project will be culminated with a set of flight-based microgravity experiments intended to validate the experimental correlations, mechanistic descriptions, and analytical models that are produced during the ground-based portion of this project and several other areas of investigation included within the SETA project. It is proposed that the planning and performance of flight-based solidification experiments will be done collaboratively, as will the analysis of post-solidification specimens upon their return to Earth.

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As mentioned in the Task Description, the objectives of this project are aligned with developing both the fundamental knowledge and practical technology to achieve controlled multiphase structures through advanced solidification processing. The potential Earth-based impact of this research is indeed far-reaching, given the extremely broad range of multiphase structures that can be fabricated through solidification processing, which is certainly not limited to metals and alloys. Indeed the properties of nearly any multiphase material can be improved by better control of multiphase structure. In addition, the fundamental and practical knowledge from this research may lead to innovative process technologies and new materials and/or structures.
Task Progress:	[Ed. note May 2017: compiled from final technical report] The rich landscape of growth morphologies that are attainable through solidification processing of ternary and higher order systems provides a promising avenue for accessing new structures, materials, and devices with enhanced functionality. However, the challenge of controlling the complex inter play between coupled-phase topology, local defect structures, and non-planar growth modes requires a more complete understanding of the mechanisms of pattern formation and the influence of applied external forces. Focusing on multiphase microstructures in Al-Cu-Ag ternary alloys, this experimental program is aimed at understanding and quantifying the 3D relationships between the relevant structural features, including local eutectic phase topology, crystallographic orientation/misorientation, eutectic fault configuration, faulted domain structure, and grain structure, all relative to the constraints of specimen geometry and applied thermal field. The project was executed between the dates of October 1, 2010 and September 30, 2015. In this final report, the overall accomplishments of the project are summarized and specific key findings are highlighted. Additional publications will be produced subsequent to this report. Important project findings and outcomes are listed below, categorized into several major technical themes. These are open areas of investigation and will continue to be studied in the follow-on project (PI: A. Genau), which is targeted for flight (ISS) experiments. GRAIN COMPETITION AND SELECTION IN A BINARY EUTECTIC • Grain selection in the binary eutectic alloys was studied within a 1.25 µm/s to 10 µm/s velocity range. The initial number of grains scales with growth velocity. Well ordered grains tend to be selected within the first 10 to 20 mm of directional growth, and order evolution is well captured by FFT analysis. The number decreases rapidly during growth, with a full grain selection transient length of 50-75 mm. Maze-like grains are observed only at very low growth rates (< 0.001 mm/s). • Eutectic grain boundary structure that forms during solidification can be quite complex, depending on crystallographic orientation and phase topology. Boundaries may be flat or highly tortuous, and the structure is related to lamellar faults. Lamellar faults are integral components of the eutectic grain boundary, and solidification in areas of topological mismatch may be a source of lamellar faults. Fault structures were observed to persist over long distances (i.e., >0.100 mm). • Boundary migration during solidification involves decoupling of the orientational and topological components of the grain boundary. Eutectic order is relatively low in the region just behind the laterally mig rating (during solidification) grain boundary. TERNARY INVARIANT STRUCTURE • The scaling of eutectic spacing with velocity is different for the different phases indicating a change in the nature of phase coupling with velocity. The geometric mean of the eutectic spacings of the three phases was used to describe the velocity dependence of the structure, and a triangular “diffusive cell” was examined as a characteristic of the coupling. The nature of three-phase and two-phase coupling and the associated diffusion pathways in the liquid is an important topic for further study. • In the velocity range 0.0005 to 0.0018 mm/s, two different morphologies are observed. Comparison with coupled growth analyses for various geometries reveals a transition in the nature of coupling. The angular component of the FFT was found to be a good indicator of the morphological transition. Simple eutectic spacing is not sufficient for differentiating growth modes. • Analysis of structure indicates a growth-mode transition at approximately 0.001 mm/s, with a brick/lamellar structure appearing at low velocities. The two-velocity (step change) experiment in which velocity increased from 0.0007 to 0.004 mm/s indicates that the change in morphology is not orientation-based but truly velocity induced. TERNARY INVARIANT POST-SOLIDIFICATION EFFECTS • The post-solidification effects are dictated by the phase diagram. The true solidification morphology can only be observed within a few millimeters of the quenched front. This structure continues to change at a given location after the solidification front advances another 15 mm ahead of that location (for G=8-9 K/mm, v=1 µm/s). • Objective measurements reveal that the morphological transition is best indicated by changes in the major caliper length and eutectic spacing of Ag2Al and the minor caliper length of Al (FCC). • A cellular automaton analysis tool was developed to obtain an estimate of the true growth front phase configuration using structures observed some known distance behind the front, using phase fraction as a control parameter. SINGLE PHASE LAYERS • In the Al-Cu binary eutectic, the single phase layer that forms during the pre-growth holding period is Al2Cu. For the ternary alloy, the pre-growth layer is a mixture of Ag2Al and Al2Cu. The development of single- and double-phase layers is an important (but poorly understood) phenomenon that affects subsequent growth. • The thermal gradient at the solid-liquid interface (controlled in these experiments by the furnace set-point) has a strong influence on the growth of single-phase layers. • For the univariant composition studied, no continuous single-phase layer formed even after 4 hours of holding.

	UNIVARIANT GROWTH AND THE DIRECTION OF GRAVITY • For the UA alloy (tested in both thick and thin tubes), the initial transient is not well predicted by a modified Smith model. The observed critical velocity for non-planar morphologies is consistent with a simple constitutional supercooling criterion (thick samples). • Comparison of inverted and standard growth experiments showed steady-state profiles measured at the quenched interface in the liquid is different for these two configurations. The profile is almost linear in the inverted configuration whereas it follows a more rapid decay in the standard configuration (gradient up, gravity down). • For the same experiments, no segregation at the 2 mm before quench is observed in the inverted specimen, whereas radial segregation is present at the same section of the standard specimen. • No single-phase layer formation is observed after a 4-hour hold in the standard configuration, but the inverted configuration results in the formation of single phase layer, although it is not continuous. NOTE: Project Leadership transitioned to Professor Amber Genau (University of Alabama, Birmingham) with project titled "Eutectic Solidification in a Ternary Alloy." See that project for subsequent reporting.
Bibliography Type:	Description: (Last Updated: 06/01/2016)
Articles in Peer-reviewed Journals	Sargin I, Genau AL, Napolitano RE. "Post-solidification effects in directionally grown Al-Ag2 Al-Al 2 Cu eutectics." Journal of Phase Equilibrium and Diffusion. 2016 Feb;37(1):75-85. http://dx.doi.org/10.1007/s11669-015-0439-6 , Feb-2016
Dissertations and Theses	Sargin I. "Invariant and univariant eutectic solidificatio in ternary alloys." Ph.D. dissertation, Iowa State University, Ames, May 2015. , May-2015